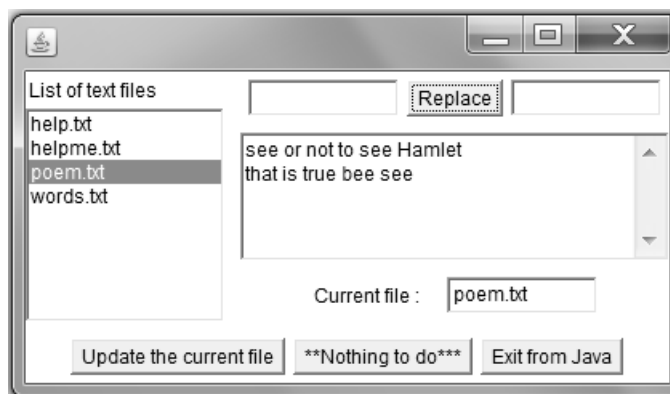
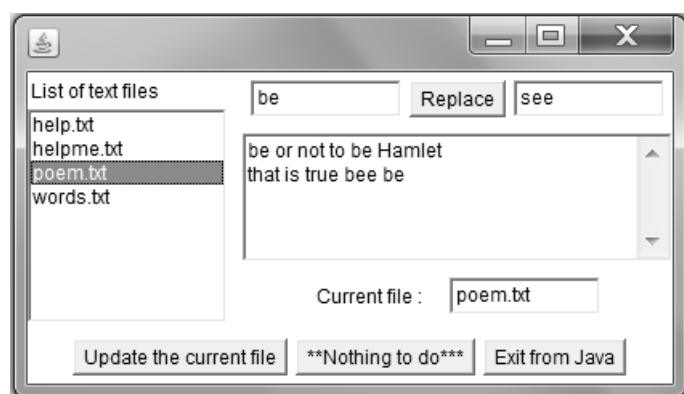


Treat the following exercises:

- I. (12 points) Consider the following interface that display within a list the names of all file with the extension ".txt" of the current folder in order to modify the contents of a selected file and save the modified contents.
The contents of a file can consist of multiple rows and each row consists of several words separated by spaces (a single space after each word).



We give you the main skeleton of the class and we ask you to complete this class:

```
import java.awt.*; import java.awt.event.*; import java.io.*;
public class Exercisel extends Frame implements ActionListener, ItemListener{
    private TextField source = new TextField(10);
    private TextField target = new TextField(10);
    private TextArea text = new TextArea(4, 24);
    private List list = new List(5);
    private Label L1 = new Label("List of text files");
    private Label L2 = new Label("Current file :");
    private TextField filename = new TextField(10);
    private Button B1 = new Button("Replace");
    private Button B2 = new Button("Update the current file");
    private Button B3 = new Button("**Nothing to do**");
    private Button B4 = new Button("Exit from Java");
    public Exercisel() {
        B1.addActionListener(this);      B2.addActionListener(this);
        B3.addActionListener(this);      B3.addActionListener(this);
        list.addItemListener(this);
        fillList();
        frameLayout();
        this.pack();      this.setVisible(true);
    }
    public void itemStateChanged( ItemEvent evt) {
        try { openFile(); }catch (IOException e) {}
    }
    public void actionPerformed (ActionEvent evt) {
        Button b = (Button) evt.getSource();
        if (b == B1) replace();
        else if (b == B2) try { update(); }catch (IOException e){}
        else if (b == B4) System.exit(0);
    }
    public static void main (String [] arg) {
        new Exercisel();
    }
}
```

More specifically, you must write the following methods:

- fillList()** : allows to find all names of the files with the extension ".txt" of the current folder and add these names to the list.
- frameLayout()** : allows to add the graphic components to the frame and to organize it, in order to obtain this graphical interface. (You must use Layouts and Panels to obtain the same interface)
- openFile()** : this method corresponds to selecting a file from the list, it allows to create a stream corresponding to the selected file and display its contents in the TextArea and its name in the TextField "Current file".
- replace()** : this method corresponds to the " Replace " button, it allows to replace all occurrences of a word of TextArea by occurrences of another word. For example (see figure) all 'be' shall be replaced by 'see'.
- update()** : this method corresponds to the " Update the current file " button, it allows to replace the contents of the selected file with the contents of the TextArea.

II. (5 points) Write a class named "Lines" which implements *Runnable* and permits to find the number of lines of a file and save this number. Write the main method, that takes during execution several arguments representing the names of files and that allows to create a thread as an instance of "Lines" for each file. The main method must wait for the end of all the threads then she must find and print the total number of lines of all the files.

III.(3 points) Write an applet program that displays an image with the actual size at the applet center and write the corresponding HTML file. The image file name is passed as parameter from the HTML file. If the image size is greater than the applet size, the message "Image can't be displayed" must appear at the applet center.

